What every sensor would read at 60 kW

SOFC POC · rated power estimate · 6 Nov 2026 · online version:
https://claude.ai/artifact/JmbcsJi74NZ4CM7YV8Vq5Z

The plant never made more than **34 kW**. We took every sensor that moved during part load and stretched its trend to the rated **60 kW**. Electrical readings and fuel stretch cleanly. The stretch also shows the stacks **cannot** make 60 kW as they are now. They would need about **44 A** per string at **0.58 V** per cell. The design point is 30 A, and the plant stops a cell below 0.72 V. At 30 A the stacks give about **51 kW**.

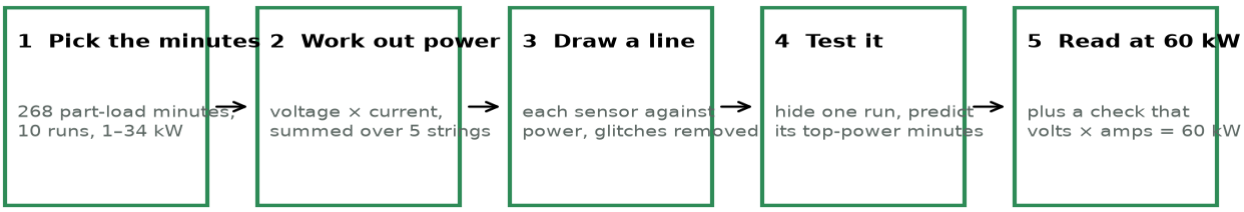


Reading	34 kW	50 kW	60 kW	Unit	Trust
Power from one string (EIRC 4.1 power)	7.1	10.6	12.7	kW	solid
Current per string (average of 5 strings)	16.7	28.8	44.1	A	solid
Voltage per string (average of 5 strings)	408	351	277	V	solid
Average cell voltage (string voltage ÷ 476 cells)	0.86	0.74	0.58	V	solid
Fuel used up (PLC fuel use, worst string)	45.9	66.2	79.0	%	solid
Natural gas flow (FIRC 1.4, from fuel chemistry)	151	180	232	ln/min	good
Stack safety temperature (TIRZ 8.1.3, trip 720 °C)	667	688	700	°C	weak
Exhaust temperature (TIRZ 8.1.10, trip 200 °C)	147	163	174	°C	weak
Cathode air flow (FIRC 1.1.1, set by operator)	4,287	4,287	4,287	ln/min	kept

If you remember one thing: the full 464-sensor table at 0–60 kW exists, but 60 kW sits past what these stacks can safely give, so treat 50–60 kW as a "what if", not as an operating point.

How the stretch works

Think of drawing a ruler line through dots on a chart, then reading the line further out than the dots go.



Follows power: the line explains at least half of the sensor's ups and downs, and with one run hidden it still predicts that run's top minutes within a quarter of the sensor's range (68 sensors).

Glitches: minutes with impossible readings (a string at 93 V, a cell at 28 V) are dropped before the final line is drawn.

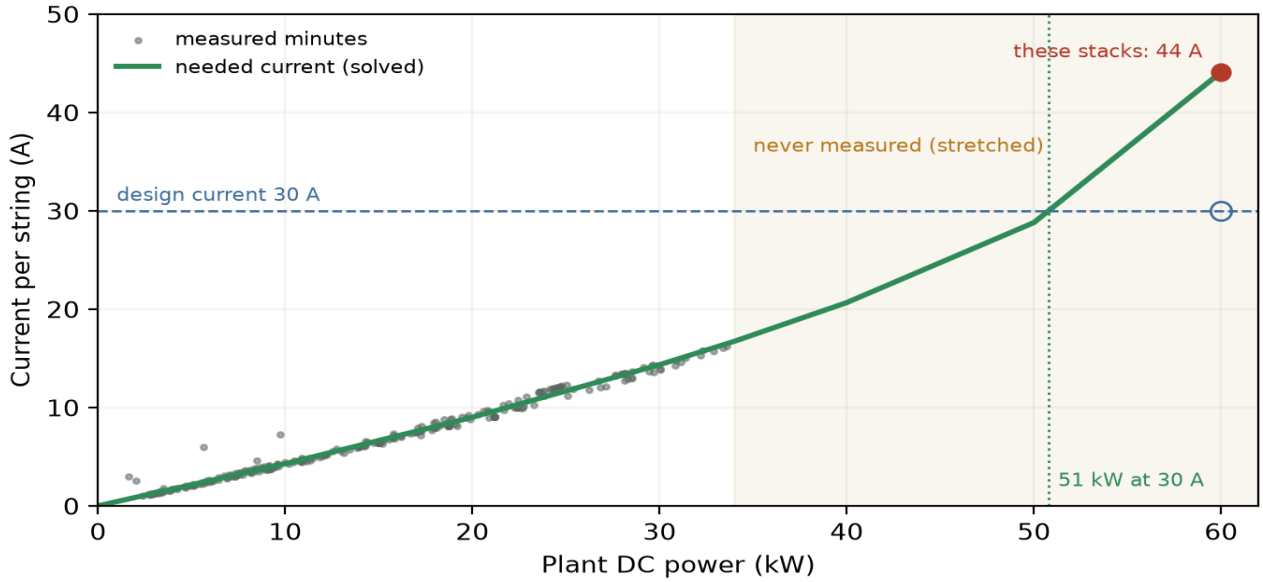
Volts × amps check: separate lines for voltage and current give only about 50 kW at "60 kW". So each string's own voltage-versus-current line is used to solve the current that gives exactly the asked power.

Gas: the operator moved gas in steps, so gas is worked out from fuel chemistry: current needs a fixed amount of methane, divided by the share of fuel used, which follows power closely.

If you remember one thing: every number is a straight-line stretch of what the plant actually did, with one physics check on the electricity.

Electricity and gas at 60 kW

The stacks lose voltage as current rises. To make up for it, 60 kW needs far more current than the design assumed.



Power kW	Current A	String V	Cell V	Fuel used %	Gas ln/min
34	16.7	408	0.86	46	151
40	20.7	389	0.82	54	160
50	28.8	351	0.74	66	180
60	44.1	277	0.58	79	232

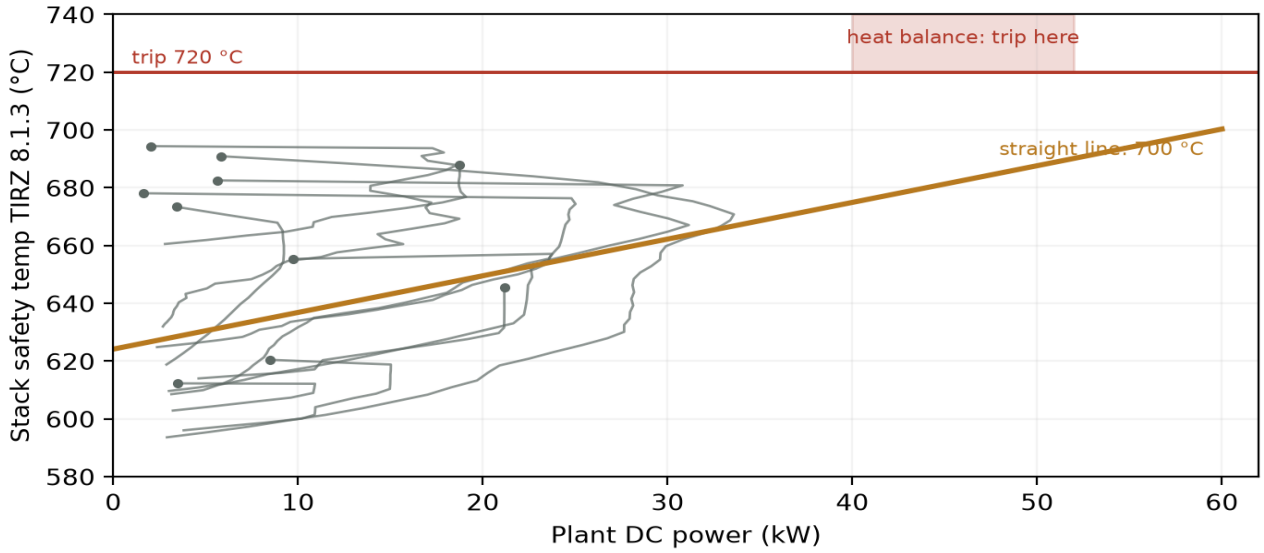
Cell voltage too low: at 60 kW the average cell sits at 0.58 V; the plant cuts load when the weakest cell falls below 0.72 V.

Gas line too small: 60 kW needs about 232 ln/min of natural gas at 79 % fuel use; the POC gas line has delivered at most about 166 ln/min.

If you remember one thing: at the design current of 30 A these stacks give about 51 kW; 60 kW would take about 44 A, past the cell limit and the gas supply.

Why temperatures do not stretch

A kettle does not boil the moment you switch it on. The stack keeps warming for hours after the power goes up, and every part-load run ended long before that.



Grey: the 10 part-load runs, dot = run end. Most runs end top-left: power already back down while the stack is at its hottest. Temperature depends on how long the load lasted, not only on how high it was.

What the table gives: for all 145 temperatures (and the lowest-cell voltages) the straight-line value, marked weak. Read it as a floor, not a forecast.

Better source for heat: the heat balance from the 5 Nov rated-power work reaches the 720 °C stack trip at 40–52 kW; exhaust hits 200 °C in 3 of 4 cases; no steady point below the trip.

Operator-set values: 251 readings are setpoints, valve modes and flows the operator chose (e.g. cathode air 3000–5000 ln/min); the table keeps their part-load average.

If you remember one thing: for heat, trust the heat balance, not the straight line: the stack trips before 60 kW with today's cooling air.

Files and decisions

03_data_analysis/scripts/rated_power_extrapolation.py → 03_data_analysis/rated_power/sensor_extrapolation.xlsx (all sensors, values at 0–60 kW), electrical_60kW.csv, string_vi_fit.csv.

Decisions

A. Top power for the synthetic rated stage: keep 60 kW as a "what if", or cap at about 51 kW (what 30 A gives). Suggested: cap at 50 kW, label above as estimated.

B. Ask the stack maker for its voltage–current curve (design 400 V per string at 30 A vs about 350 V measured trend).

C. Feed the temperature rows from the twin heat balance over time instead of the straight line.

D. Gas supply: 60 kW needs about 232 ln/min against about 166 available. Confirm the real line limit.

If you remember one thing: the table is ready to use; the open choice is whether "rated" means the design 60 kW or the roughly 50 kW these stacks can actually give.